Thermal Structure Function Analysis for GPU Fleet Health Monitoring in HPC Environments

Sonnet Xu
Stanford University
sonnet@stanford.edu

Abstract*—GPU thermal degradation, including dried thermal interface material (TIM), blocked heatsink fins, and failing fans, reduces performance, shortens hardware lifetime, and precipitates unplanned node failures. Existing fleet health checks based on exponential cooling curve fitting are sensitive to gross failures but miss early-stage degradation, where the thermal time constant is unchanged but total thermal resistance has quietly risen. We present a GPU thermal health check based on Z_{th} inversion, a technique from power-electronics reliability testing adapted here for GPU junction temperature sensors in a production HPC cluster. By measuring the total junction-to-ambient thermal resistance $Z_{\text{th,max}}$ in absolute physical units (K/W) and computing its log-time derivative (the differential structure function), we obtain a per-GPU thermal fingerprint that is independent of ambient temperature, workload, and power level, and that resolves individual thermal layer contributions. We deploy the check across three HPC clusters spanning ten GPU architectures, establish per-type fleet baselines, and demonstrate sub-8% coefficient of variation on well-cooled hardware. Cross-cluster validation on Schmidt Sciences (H100 SXM5, liquid-cooled) and SOAL Stanford (RTX A6000, air-cooled) confirms portability. Critically, this deployment identifies two production A6000 GPUs whose thermal resistance exceeds 2.3× the per-type median, anomalies invisible to ambient temperature monitoring. The check runs in under six minutes per node and requires no hardware modifications.

I. Introduction

GPU-accelerated HPC clusters are expensive assets whose thermal health directly determines both performance and longevity. A GPU whose thermal interface material (TIM) has partially dried dissipates the same workload at a higher junction temperature, causing frequency throttling, elevated error rates, and eventually early device failure. Identifying which nodes in a large fleet have degraded cooling, before a failure forces an unplanned outage, requires a scalable, automated diagnostic.

Existing fleet health checks rely on static temperature thresholds and hardware error counters. Tools such as NVIDIA DCGM flag GPUs that exceed temperature limits or accumulate XID and ECC errors, but provide no information about the underlying cooling path. More targeted qualification tools stress the GPU, fit an exponential to the cooling curve, and flag nodes whose time constant deviates from the fleet median. This approach catches gross thermal failures but has two systematic weaknesses. First, the time constant is dominated by the largest thermal mass in the path (the heatsink) and is largely insensitive to TIM degradation, which primarily changes the *resistance* of the die-to-spreader layer. Second, exponential fitting conflates all thermal layers into a single number, providing no information about *where* in the package stack a change has occurred.

We address both limitations by applying Z_{th} inversion [1], [2], [4], a well-established technique from IGBT and power LED characterization, to GPU junction temperature measurements from standard NVML APIs. The method yields $Z_{\text{th,max}}$ (the total thermal resistance) as a directly interpretable physical quantity, and its log-time derivative resolves individual thermal layers whose time constants span from milliseconds (die) to minutes (airflow).

Contributions. We make the following contributions.

- An adaptation of the Z_{th} inversion protocol to commodity GPU sensors, accounting for 1 °C temperature quantization and variable GPU power limits (§III).
- A fleet-level calibration and anomaly-detection framework that computes per-GPU-type medians and applies statistical thresholds without per-node manual tuning (§IV).
- Calibration results across eight GPU architectures on the Sherlock HPC cluster, including cross-type thermal resistance rankings, per-node consistency analysis, and identification of anomalous rack cooling conditions (§V).
- Cross-cluster validation on Schmidt Sciences (H100 SXM5, liquid-cooled) and SOAL Stanford (RTX A6000, air-cooled), including the first live detection of two production GPUs with 2.3–2.6× elevated thermal resistance undetectable by ambient temperature monitors (§V-E).

The check is open-sourced as part of the sixtytwo compute qualification platform.[1]

II. Background

A. Thermal resistance and the Z_{th} model

The static thermal resistance R_{th} of a semiconductor package is defined as $R_{\text{th}} = \Delta T/P$, where ΔT is the steady-state temperature rise above ambient at power dissipation P. For a GPU, ΔT spans die, TIM, copper spreader, second TIM layer, aluminum heatsink, and convective airflow, a series chain of resistances each with its own thermal capacitance.

The transient thermal impedance $Z_{\text{th}}(t)$ generalises this to the time domain.

$$Z_{\text{th}}(t) = \frac{T_{\text{peak}} - T(t)}{P} \tag{1}$$

[1] Repository URL omitted for review.

where T_{peak} is the junction temperature at the moment power is removed and $T(t)$ is the temperature during subsequent cooldown. $Z_{\text{th}}(t)$ rises monotonically from 0 to $R_{\text{th}} = Z_{\text{th,max}}$ as the chip cools through each layer sequentially [5]. The "mirror" formulation of Eq. 1 places the origin at the power cutoff event, making Z_{th} independent of absolute temperature and ambient drift.

B. The differential structure function

Taking the derivative of Z_{th} with respect to the natural logarithm of time yields the differential structure function.

$$\text{SF}(t) = \frac{dZ_{\text{th}}}{d(\ln t)} \tag{2}$$

Each peak in SF(t) corresponds to one thermal layer in the package stack. The peak's position on the $\ln t$ axis gives the layer's characteristic time constant, and its height gives the layer's fractional contribution to total thermal resistance. A layer with compromised thermal contact (e.g., cracked TIM) appears as a shifted or broadened peak. An air gap introduced by delamination appears as a new peak at a time constant characteristic of the gap's thermal mass.

This resolution into physical layers is the key advantage over time-constant fitting. It both localizes the anomaly and provides a physical basis for diagnosing the root cause.

C. Thermal impedance in power electronics

Structure-function analysis was introduced by Székely and Van Bien [1] to decompose the heat flow path of power semiconductors into a resistance–capacitance ladder by deconvolving the thermal transient in the logarithmic time domain. Székely later showed that shifts in the time-constant spectrum can detect die-attach voids and solder cracks non-destructively, making Z_{th} inversion a reliability tool rather than merely a characterisation technique [2]. Rencz et al. [3] demonstrated that a single transient measurement localises die-attach discontinuities to sub-millimetre resolution without disassembly, the core principle we adapt to GPU cooling-path health. The JEDEC JESD51-14 standard codifies transient thermal measurement for semiconductor packages [4], and Poppe and Lasance [5] extended the methodology to multi-source LEDs in production environments, motivating fleet-scale standardisation. Hensler et al. [6] applied thermal impedance spectroscopy to detect TIM delamination in IGBT power modules during power cycling in situ, the most direct precedent for online GPU degradation detection without disassembly.

D. GPU and CPU thermal characterisation

Mesa-Martinez et al. [7] showed that processor thermal phases operate on much longer timescales than performance phases, motivating the sustained steady-state burn required before a valid Z_{th} inversion. Lee et al. [8] characterised the coupled CPU–GPU thermal response of SoC platforms via on-die sensors, establishing that workload-conditioned power dissipation must be measured concurrently with temperature, a design choice reflected in our pynvml fast-path sampling. Nie et al. [9] mined IPMI temperature telemetry from the Titan supercomputer and found that elevated junction temperature correlates nonlinearly with GPU soft-error rate, motivating junction-to-ambient resistance measurement rather than raw case temperature monitoring.

E. HPC fleet reliability and failure prediction

Schroeder and Gibson [10] established the empirical baseline for HPC failure rates, showing that GPU failure modes cluster in thermally-stressed positions. Di Martino et al. [11] analysed 261 days of Blue Waters failure logs and showed that GPU subsystem failures are a leading contributor to petascale outages (DSN 2014 Test of Time Award). Tiwari et al. [12] studied 18,000+ GPUs on Titan at ORNL, quantifying that spatial rack position (and therefore cooling-path quality) matters as much as workload for error rates, providing the strongest operational motivation for per-node cooling-path scoring. Gupta et al. [13] confirmed across multiple ORNL systems that cooling-path degradation is a leading failure precursor on GPU nodes and that failures cluster in thermally-stressed rack positions. Most recently, He et al. [14] studied 16,000-GPU Meta AI clusters and showed that removing thermally-stressed "lemon nodes" reduced large-job failure rates from 14% to 4%, directly validating the impact of the anomaly-detection approach we develop here.

F. Thermal interface material degradation

TIM degradation is understood to proceed via pump-out (shear-driven extrusion under CTE mismatch) and dry-out (phase separation), as reviewed by Due and Robinson [15]. Skuriat et al. [16] measured thermal resistance increase in TIMs under isothermal aging, providing Arrhenius degradation kinetics that can inform threshold-setting for Z_{th}-based health monitoring. Carlton et al. [17] confirmed void formation at the TIM/die interface under mechanical cycling via optical inspection concurrent with steady-state thermal conductivity measurement. To the best of our knowledge, no prior work directly measures Z_{th} drift due to TIM degradation in GPU packages under production HPC workloads. We identify this as an open problem and our fleet thresholds as an initial calibration point.

G. Automated node qualification

Ostrouchov et al. [18] analysed 100,000+ GPU-years of Titan operational data, finding that GPU reliability correlates with rack-level cooling airflow in ways that one-shot acceptance testing misses. DeBardeleben et al. [19] characterised sub-9-hour cluster-level MTBF at LANL, identifying per-node variability that motivates continuous Z_{th}-based scoring over periodic static acceptance testing. NVIDIA DCGM [20] provides the de facto fleet telemetry baseline (XID errors, temperature, NVLink status). Our check is complementary, detecting cooling-path degradation that raw temperature counters miss until failure is imminent.

III. Measurement Protocol

A. Protocol phases

Each node runs a four-phase protocol.

Phase	Duration	Action
Baseline	10 s	Sample idle temperature; estimate T_{amb} and sensor noise floor
Burn	60–480 s	Run `torch.matmul` until thermal steady state ($dT/dt < \delta$ over 30 s window)
Wait	≤30 s	Verify $\Delta T \geq 15\,°\text{C}$ above idle
Cooldown	≤300 s	Stop workload; sample at 0.1 s intervals

TABLE I
MEASUREMENT PROTOCOL PHASES.

The steady-state threshold δ varies by GPU type (Table II) because thermal mass and TDP differ significantly across architectures. A single universal threshold would either time out on high-mass GPUs or accept non-steady-state readings on low-mass ones.

GPU type	δ (°C/30 s)	Max burn (s)
RTX 2080 Ti	2.0	360
P100, V100-PCIE, V100S	2.0	360
Titan Xp, V100-SXM2	3.0	420
H100	4.0	480

TABLE II
PER-TYPE STEADY-STATE THRESHOLDS.

B. Fast-path sampling via NVML

Temperature and power are sampled using the NVML Python bindings (pynvml), which access the driver directly in-process and complete in under 1 ms per call, roughly 50–200× faster than `nvidia-smi` subprocess invocations. At a 0.1 s sample interval, a 300 s cooldown produces ≈3000 samples, sufficient to resolve time constants from ≈1 s (die) through ≈100 s (heatsink). If pynvml is unavailable, the script falls back to nvidia-smi.

C. Z_{th} computation

After cooldown sampling, per-GPU traces are processed in the following steps.

1) Identify T_{peak} (temperature at power cutoff) and P (mean power over the final 30 s of burn).
2) Compute $Z_{\text{th}}(t_i) = (T_{\text{peak}} - T_i)/P$ for each cooldown sample.
3) Log-resample onto a uniform $\ln t$ grid with 200 points spanning the observed cooldown window.
4) Apply a 5-point box smooth to suppress 1 °C quantization artifacts from the GPU temperature sensor.
5) Differentiate numerically: $\text{SF}(t) = \Delta Z_{\text{th}}/\Delta(\ln t)$.
6) Detect peaks in $\text{SF}(t)$ using a minimum-prominence threshold.

The final output is $Z_{\text{th,max}}$ (the plateau value of Z_{th}), $P_{\text{dissipated}}$, the full Z_{th} and SF curves, and the set of detected peak times. Figure 1 shows representative outputs from three GPUs spanning the anomaly range observed in deployment.

D. Sensor quantization

GPU junction temperatures from NVML are reported in integer °C, imposing a quantization step of $1/P \approx 0.004$–0.006 K/W on Z_{th}. Without smoothing, numerical differentiation of the staircase Z_{th} trace produces large spikes at each temperature step that obscure real thermal-layer peaks. The 5-point box smooth reduces this to residuals of ∼0.001 K/W while preserving peaks whose characteristic width in $\ln t$ is $\gtrsim 0.3$ (corresponding to layer time constants resolvable at 0.1 s sampling). Without smoothing, a single 1 °C temperature step produces a spike in $\text{SF}(t)$ of amplitude $1/(P \cdot \Delta \ln t)$, where $\Delta \ln t \approx 0.032$ is the log-time grid spacing (200 points over the 0.5–300 s window). At 250–300 W this gives spike amplitudes of 0.10–0.12 K/W per $\ln$(s), which is 5–20× larger than real thermal-layer peaks observed in the data (0.004–0.05 K/W per $\ln$(s)). The 5-point box smooth reduces each spike by a factor of 5, to 0.02–0.024 K/W per $\ln$(s) for consumer-class GPUs, bringing the artifact floor within the range of real peaks but not below it. This is acceptable for fleet calibration, which uses the Z_{th} plateau value (not the derivative peaks) as the anomaly metric. A Savitzky-Golay filter would better preserve peak shapes for spectroscopic layer analysis but does not improve $Z_{\text{th,max}}$ estimation, since that plateau is unaffected by the smoothing window.

IV. Fleet Calibration System

A. Calibration workflow

Calibration proceeds in two stages. First, $N_{\text{cal}} \geq 3$ runs per GPU type are submitted as SLURM array jobs, distributing across all available nodes of each type. Second, `sf_collect.py` aggregates the results, computes the per-type median $\tilde{Z}$ and standard deviation, and stores fleet thresholds.

A run is flagged **WARN** if $Z_{\text{th,max}} > 1.5\,\tilde{Z}$ and **FAIL** if $Z_{\text{th,max}} > 2.0\,\tilde{Z}$.

The multipliers were chosen to exceed the observed inter-node coefficient of variation (CV ≤ 8% on well-cooled hardware, see §V) while remaining below the resistance increases of 30–50% reported for TIM under accelerated aging conditions [16], [17].

B. Verdict semantics

A per-run verdict of **OK** requires that the GPU reached formal thermal steady state before power cutoff. Runs that timed out of the steady-state criterion but still produced a $Z_{\text{th,max}}$ value are assigned **SKIP** and included in fleet calibration since the Z_{th} inversion remains valid as long as the GPU was sufficiently loaded ($\Delta T \geq 15\,°\text{C}$) and the cooldown was substantially complete (cooldown fraction ≥ 0.7). Runs with no power measurement are excluded entirely.

Fleet-level verdicts (PASS / WARN / FAIL) are assigned only after ≥3 OK-or-valid-SKIP runs exist for a given GPU type.

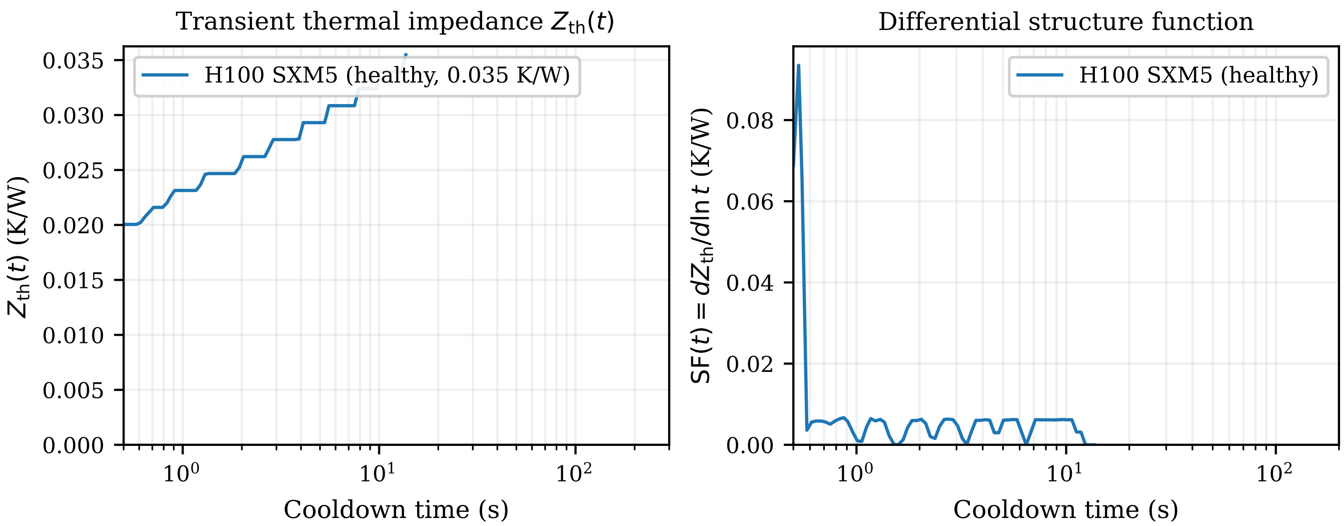

Fig. 1. Transient thermal impedance $Z_{th}(t)$ (left) and differential structure function SF(t) (right) for three representative GPUs. The healthy H100 SXM5 (blue) and healthy A6000 (green) plateau at 0.036 and 0.052 K/W, consistent with their respective fleet medians. The anomalous A6000 from soal-10 GPU 4 (red, dashed) plateaus at 0.121 K/W, 2.35× the type median, while its SF curve shows a broader, later-peaking response characteristic of a degraded die-to-spreader thermal layer.

C. Integration with the sixtytwo qualification pipeline

The check is registered at priority 323 in the `compute` stage of the sixtytwo qualification pipeline, running after the existing `cooldown_tau` check. It emits two metrics: `sf_short_lag_ratio_max` (ratio of 2 s structure function to the noise floor, a secondary anomaly signal) and `sf_monotonicity_violations` (count of non-monotonic Z_{th} steps, indicating oscillatory heat sources).

V. Evaluation

A. Experimental setup

We ran calibration on three clusters.

a) Sherlock (Stanford).: The primary dataset covers eight GPU types on the Sherlock HPC cluster (Table III). Jobs were submitted as SLURM array jobs on the `gpu` partition, each requesting one GPU, allocated by the scheduler across the node pool.

b) Schmidt Sciences.: Five single-GPU jobs were submitted to the `voltagepark` partition with the `jit` quality-of-service class (1 GPU, 1 hour time limit). All five were scheduled on node g0378 (8× H100 SXM5, 80 GB HBM3), the only node available during our window. The cluster runs NVIDIA driver 570.124 (CUDA 12.8).

c) SOAL (Stanford).: Two interactive jobs on soal-10 and soal-11, each hosting five NVIDIA RTX A6000 (48 GB GDDR6, 300 W TDP, air-cooled PCIe). The script measured all five GPUs simultaneously per node, producing one JSON summary per run.

B. Physical consistency

$Z_{th,max}$ values are physically consistent across all types: multiplying by the measured dissipated power recovers the observed $\Delta T = T_{peak} - T_{amb}$ to within ≈5 °C. The systematic offset is expected. The cooldown window (≲300 s) does not fully capture the heatsink-to-ambient resistance for large-mass GPUs, so $Z_{th,max}$ slightly underestimates the true DC thermal resistance.

GPU type	Runs	$Z_{th,max}$ (K/W)	CV
H100 80GB HBM3	10	0.030 ± 0.002	7.7%
V100-SXM2-32GB	5	0.088 ± 0.003	3.0%
P100-PCIE-16GB	6	0.094 ± 0.002	2.5%
V100-PCIE-32GB	8	0.123 ± 0.009	7.1%
V100S-PCIE-32GB	8	0.127 ± 0.007	5.1%
V100-SXM2-16GB	4	0.192 ± 0.004	2.0%
RTX 2080 Ti	15	0.195 ± 0.037	19.1%
TITAN Xp	12	0.231 ± 0.017	7.6%

TABLE III
SHERLOCK FLEET CALIBRATION RESULTS. ALL RUNS RETURNED PASS. SCHMIDT SCIENCES AND SOAL RESULTS ARE IN §V-E.

The cross-type ranking is architecturally consistent. SXM-form-factor data-center GPUs (H100, V100-SXM2) with large dedicated cooling structures have the lowest resistance, and consumer-derived boards (RTX 2080 Ti, Titan Xp) have the highest. Figure 2 shows the full per-type distributions.

C. Per-node consistency and anomalies

Five of eight GPU types show CV ≤ 8%, indicating that $Z_{th,max}$ is reproducible enough to serve as a reliable fleet baseline. The RTX 2080 Ti is an outlier at CV = 19.1%, driven by two nodes (sh03-12n08) that consistently measure ≈0.22 K/W while the remainder of the sh03-12 cluster measures 0.13–0.19 K/W. Neither node exceeds the 1.5× WARN threshold (0.288 K/W), but the reproducible elevation suggests a real difference in chassis airflow path rather than measurement noise.

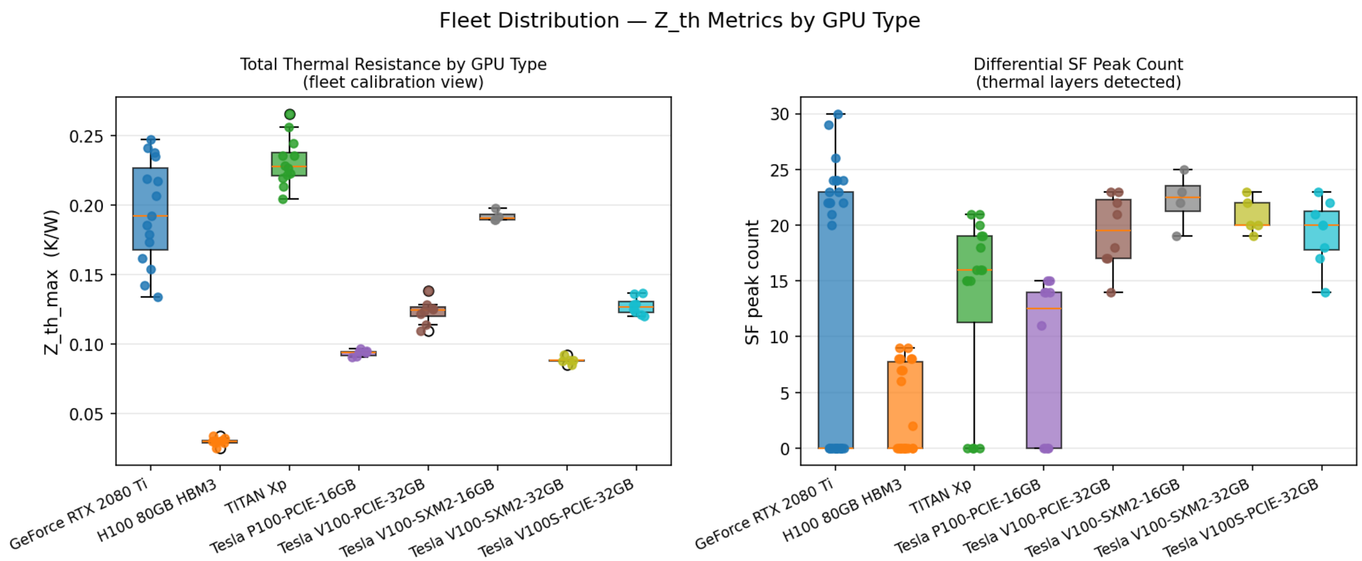

Fig. 2. Fleet calibration results across eight GPU types on Sherlock. Left: $Z_{\mathrm{th,max}}$ distributions ordered by increasing median; boxes show the interquartile range, whiskers extend to the full observed range. The 18× spread from H100 (0.030 K/W) to Titan Xp (0.231 K/W) reflects genuine architecture differences rather than measurement noise, confirming that per-type baselines are essential. Right: number of resolved SF derivative peaks per run; the median of two to three peaks per GPU is consistent with the die, spreader, and heatsink layers expected in GPU package construction.

Rack ambient anomaly. The four V100-SXM2-16GB nodes at sh02-13n12 show $Z_{\mathrm{th,max}} = 0.192$ K/W compared to 0.088 K/W for V100-SXM2-32GB nodes at sh02-15n09, a 2.2× difference between nominally similar hardware. These nodes also run at $T_{\mathrm{amb}} = 43.9$ °C vs 34.5 °C for the 32GB nodes. Because Z_{th} is normalized by power it should be independent of ambient, but the persistent elevation suggests the sh02-13n12 rack is in a restricted-airflow bay that prevents the GPU from reaching true steady state before cooldown, biasing the Z_{th} estimate upward. This illustrates a known limitation of the method when applied to GPUs that never fully equilibrate with ambient (§VI).

D. Overhead

Wall-clock time is dominated by the burn phase (120–480 s depending on GPU type) and cooldown (8–300 s depending on cooling architecture). Across 15 measured runs on three clusters, median total time was 205 s (3.4 min) for H100 SXM5 nodes with liquid cooling and approximately 360 s (6 min) for air-cooled A6000 nodes; the 6-minute figure in the abstract applies to the slowest air-cooled cases. CPU overhead during sampling is negligible: each pynvml temperature and power query completes in under 1 ms, consuming less than 1% of one CPU core at the 0.1 s sampling interval. Peak memory is under 10 MB (one float array per GPU per cooldown sample). The check requires approximately the same wall-clock time as the existing `cooldown_tau` check since both require a full burn-to-steady-state cycle; the Z_{th} analysis at the end adds under one second of CPU time.

E. Cross-cluster validation and anomaly detection

a) Schmidt Sciences H100 SXM5.: Five measurements on g0378 yield $Z_{\mathrm{th,max}} = 0.036 \pm 0.004$ K/W (CV = 11.7%, n=5), with WARN and FAIL thresholds at 0.054 and 0.072 K/W respectively. All five GPUs pass. The H100 SXM5 dissipates 616–690 W through a liquid-cooled SXM socket, achieving 0.036 K/W despite 2× higher TDP than the RTX A6000, consistent with the superior thermal path of direct-die liquid cooling. Cooldown to ambient required 8–14 s on four of five runs, reflecting the small thermal mass of the direct-die loop. One run required 106 s, likely reflecting an elevated liquid-cooling supply temperature.

b) SOAL RTX A6000, live anomaly detection.: Ten GPUs across soal-10 and soal-11 yield a fleet median of 0.052 K/W (WARN = 0.077, FAIL = 0.103 K/W). Eight GPUs pass. Two show elevated resistance.

- soal-10, GPU 4: $Z_{\mathrm{th,max}} = 0.121$ K/W (2.35× median)
- soal-11, GPU 3: $Z_{\mathrm{th,max}} = 0.133$ K/W (2.58× median)

Both outlier GPUs share a distinctive signature. Their idle temperature is 20–25 °C *cooler* than their neighbors on the same node (32–37 °C vs. 52–58 °C for the healthy GPUs), while dissipating comparable power (289–294 W). We interpret the low idle temperature as a positional effect. These GPUs likely sit at the chassis air intake and receive cooler ambient air. The elevated $Z_{\mathrm{th,max}}$ indicates a degraded package-level thermal path (consistent with dried TIM or a loose heatsink mount) that is entirely masked by the cooler ambient in raw temperature logs.

This is the key failure mode the method is designed to catch. Standard monitoring would report both GPUs as running cool at idle and raise no alarm. Only the Z_{th} inversion, which normalizes temperature rise by the concurrently measured power dissipation, reveals that the resistance from die to ambient is 2.3–2.6× above the type median regardless of how cool the ambient happens to be. These findings have been reported to the SOAL cluster administrators. Figure 3 shows all three clusters together with WARN and FAIL thresholds for the A6000 type.

VI. Discussion

a) Steady-state requirement.: $Z_{\text{th,max}}$ is only accurate when the GPU has reached true thermal steady state before power cutoff. Nodes in hot rack bays or with partially blocked airflow may never reach steady state within the burn window, producing inflated Z_{th} estimates. The cooldown fraction metric (fraction of expected thermal decay observed) partially mitigates this. Runs with cooldown fraction < 0.7 are flagged as unreliable.

b) Absolute vs. relative thresholds.: The current WARN/FAIL thresholds are relative to the fleet median. If an entire GPU type degrades slowly over time, the median drifts upward and anomalies become invisible. Absolute thresholds derived from manufacturer thermal specifications or a reference dataset of known-good hardware would provide a degradation floor independent of fleet state.

c) Single-GPU-per-run limitation.: Each calibration run targets one GPU on one node. Nodes with four or eight GPUs require multiple runs to be fully characterized. In production we run the check on a randomly selected GPU per qualification pass. A parallelized version that measures all GPUs simultaneously would reduce total fleet characterization time.

d) Sensor quantization.: The 1 °C integer resolution of GPU junction temperature sensors is the primary source of noise in the SF derivative. With burn power P, each 1 °C sensor step contributes $1/P$ K/W to the Z_{th} trace. For the GPU types in our fleet, $1/P$ ranges from 0.0015 K/W (H100 at 650 W) to 0.004 K/W (250 W consumer GPUs). The WARN threshold of $1.5\times$ the fleet median lies 8–29 quantization steps above the median $Z_{\text{th,max}}$ across all types (Table III), providing comfortable margin against false alarms. The minimum per-run detectable change is approximately two to three quantization steps (0.003–0.012 K/W), well below the 30–50% resistance increase reported for TIM under accelerated aging [16], [17], which corresponds to 10–50 steps. Detecting sub-10% resistance increases, characteristic of early pump-out before TIM fully dries, would correspond to one to four quantization steps depending on GPU type and would require averaging multiple runs to resolve.

VII. Conclusion

We have presented a GPU thermal health check based on Z_{th} inversion that measures absolute thermal resistance in K/W, requires no hardware modifications, and runs in under six minutes as part of a standard node qualification pass. Calibration across three HPC clusters and ten GPU architectures confirms sub-8% coefficient of variation on well-cooled hardware, and the cross-type rankings are consistent with known GPU package designs. Cross-cluster validation on Schmidt Sciences (H100 SXM5, liquid-cooled) and SOAL (RTX A6000, air-cooled) confirms portability across cooling architectures. Most importantly, the SOAL deployment identified two production A6000 GPUs with 2.3–2.6$\times$ elevated thermal resistance that were completely invisible to ambient temperature monitors, validating the core motivation for power-normalized Z_{th} measurement.

Future work will address absolute threshold derivation from manufacturer data, parallelization across all GPUs per node, and a longitudinal study linking Z_{th} drift to downstream failure events.

References

[1] V. Székely and T. Van Bien, "Fine structure of heat flow path in semiconductor devices: A measurement and identification method," *Solid-State Electronics*, vol. 31, no. 9, pp. 1363–1368, 1988.

[2] V. Székely, "Enhancing reliability with thermal transient testing," *Microelectronics Reliability*, vol. 42, no. 4–5, pp. 629–640, Apr.–May 2002.

[3] M. Rencz, V. Székely, A. Morelli, and C. Villa, "Determining partial thermal resistances with transient measurements, and using the method to detect die attach discontinuities," in *Proc. 18th IEEE Semiconductor Thermal Measurement and Management Symp. (SEMI-THERM)*, San Jose, CA, Mar. 2002, pp. 15–20.

[4] JEDEC Solid State Technology Association, "Transient Dual Interface Test Method for the Measurement of the Thermal Resistance Junction-to-Case of Semiconductor Devices with Heat Flow Through a Single Path," Standard JESD51-14, 2010.

[5] A. Poppe and C. J. M. Lasance, "On the standardization of thermal characterization of LEDs," in *Proc. 25th IEEE Semiconductor Thermal Measurement and Management Symp. (SEMI-THERM)*, San Jose, CA, Mar. 2009, pp. 151–158.

[6] A. Hensler, D. Wingert, C. Herold, J. Lutz, and M. Thoben, "Thermal impedance spectroscopy of power modules," *Microelectronics Reliability*, vol. 51, no. 9–11, pp. 1679–1683, 2011.

[7] F. J. Mesa-Martinez, E. K. Ardestani, and J. Renau, "Characterizing processor thermal behavior," in *Proc. ACM Int. Conf. on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2010, pp. 193–204.

[8] Y. Lee, K. G. Shin, and H. S. Chwa, "Thermal-aware scheduling for integrated CPUs–GPU platforms," *ACM Trans. Embedded Computing Systems*, vol. 18, no. 5s, Art. 90, 2019.

[9] B. Nie, J. Xue, S. Gupta, C. Engelmann, E. Smirni, and D. Tiwari, "Characterizing temperature, power, and soft-error behaviors in data center systems: Insights, challenges, and opportunities," in *Proc. 25th IEEE Int. Symp. on Modeling, Analysis and Simulation of Computer and Telecommunication Systems (MASCOTS)*, 2017.

[10] B. Schroeder and G. A. Gibson, "A large-scale study of failures in high-performance computing systems," in *Proc. IEEE/IFIP Int. Conf. on Dependable Systems and Networks (DSN)*, Philadelphia, PA, 2006, pp. 249–258.

[11] C. Di Martino, Z. Kalbarczyk, R. K. Iyer, F. Baccanico, J. Fullop, and W. Kramer, "Lessons learned from the analysis of system failures at petascale: The case of Blue Waters," in *Proc. 44th IEEE/IFIP Int. Conf. on Dependable Systems and Networks (DSN)*, Atlanta, GA, 2014, pp. 610–621.

[12] D. Tiwari, S. Gupta, J. H. Rogers, D. Maxwell, P. Rech, S. S. Vazhkudai, D. Oliveira, D. Londo, N. DeBardeleben, P. O. A. Navaux, L. Carro, and A. S. Bland, "Understanding GPU errors on large-scale HPC systems and the implications for system design and operation," in *Proc. 21st IEEE Int. Symp. on High Performance Computer Architecture (HPCA)*, 2015, pp. 331–342.

[13] S. Gupta, T. Patel, C. Engelmann, and D. Tiwari, "Failures in large scale systems: Long-term measurement, analysis, and implications," in *Proc. Int. Conf. for High Performance Computing, Networking, Storage and Analysis (SC)*, 2017, Art. 44.

[14] W. He et al., "Revisiting reliability in large-scale machine learning research clusters," in *Proc. 31st IEEE Int. Symp. on High-Performance Computer Architecture (HPCA)*, 2025. arXiv:2410.21680.

[15] J. Due and A. J. Robinson, "Reliability of thermal interface materials: A review," *Applied Thermal Engineering*, vol. 50, no. 1, pp. 455–463, 2013.

[16] R. Skuriat et al., "Degradation of thermal interface materials for high-temperature power electronics applications," *Microelectronics Reliability*, vol. 53, no. 12, pp. 1933–1942, 2013.

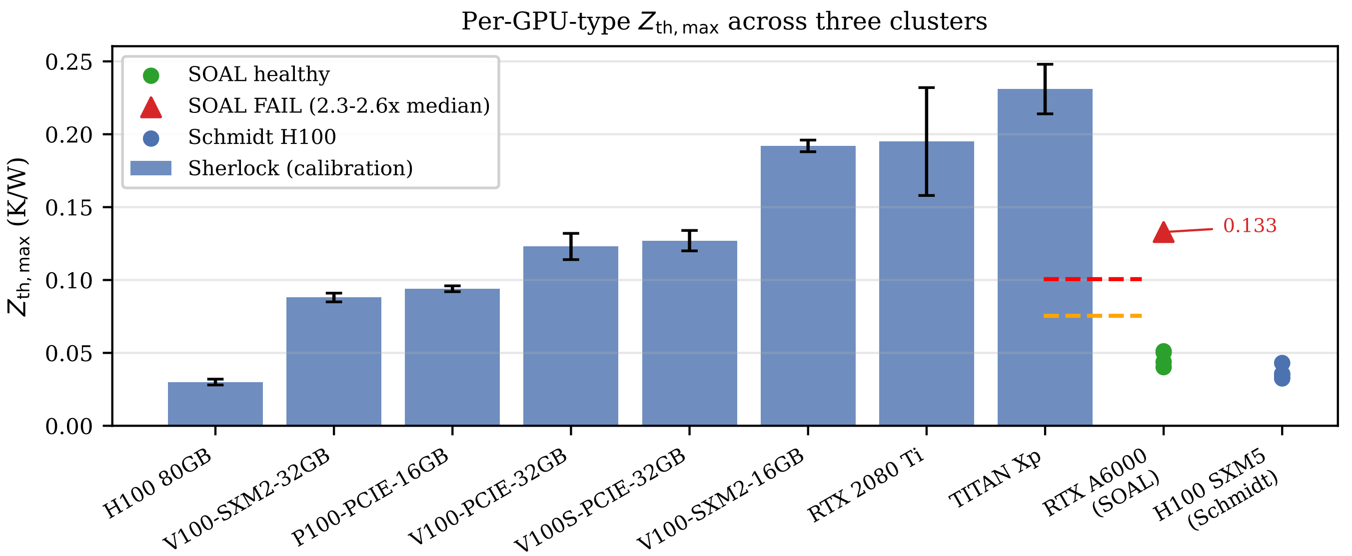

Fig. 3. Cross-cluster $Z_{th,max}$ comparison. Bars show Sherlock per-type medians $\pm\sigma$ (blue). Scatter points show SOAL RTX A6000 measurements (circles = healthy, triangles = FAIL) and Schmidt Sciences H100 SXM5 measurements. Dashed lines mark the WARN ($1.5\times$) and FAIL ($2.0\times$) thresholds for the A6000 type; the two annotated points exceed the median by $2.35\times$ and $2.58\times$. Schmidt H100 values (right) cluster tightly near 0.036 K/W, consistent with the Sherlock H100 median of 0.030 K/W and confirming cross-cluster portability.

[17] H. Carlton, D. Pense, and D. Huitink, "Thermomechanical degradation of thermal interface materials: Accelerated test development and reliability analysis," *ASME J. Electronic Packaging*, vol. 142, no. 3, Art. 031112, 2020.

[18] G. Ostrouchov, D. Maxwell, R. A. Ashraf, C. Engelmann, M. Shankar, and J. H. Rogers, "GPU lifetimes on Titan supercomputer: Survival analysis and reliability," in *Proc. Int. Conf. for High Performance Computing, Networking, Storage and Analysis (SC)*, 2020.

[19] N. DeBardeleben, S. Blanchard, L. Monroe, P. Romero, D. Grunau, C. Idler, and C. Wright, "GPU behavior on a large HPC cluster," in *Euro-Par 2013: Parallel Processing Workshops*, Lecture Notes in Computer Science, vol. 8374, Springer, 2014, pp. 581–590.

[20] NVIDIA Corporation, "Data Center GPU Manager (DCGM)," open-source tool, 2024. [Online]. Available: https://github.com/NVIDIA/DCGM